

ANTOINE DESJARDINS

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Senior Managing Data Scientist at IBM driven by the use of AI for scientific applications and the development of ML teams. Areas of interest are AI for Healthcare and AI for Science.

EXPERIENCE

09/2024 – 02/2026

EXPERT DATA SCIENTIST, IBM CLIENT INNOVATION CENTER

As the leader of the Paris Skill Center, I have the responsibility to grow the team of IBMers, for the Science Next team which I'm a part of. Our goal is to revolutionize R&D using **foundation models for science**. We are a small, start-up-like team within IBM Consulting with high ownership. My main project is to train a LLM-like model for chemical formulation for l'Oréal.

09/2024 – 02/2026

HEAD OF AI, ANEO

As the head of the AI practice of this consulting firm, I grew and lead the AI team from 6 to 11 members. I drafted proposals and won several projects, while developing internal research projects in partnership with the BOPA chair of APHP (liver steatosis prediction) and the University of Luxemburg (building the MLFlow of adversarial ML). Some achievements highlights are:

- Built a **MLOps** infrastructure for 300+ Quant Researchers/developers in a hedge fund
- Lead AI strategy for Pathé group (Theaters and Film production), reporting to CxOs
 - Tech Lead for an **attendance forecasting** ML project with a scope of ~2000 screens
 - Drove up usage of internal GenAI applications to 300+ monthly users
 - Oversaw a portfolio of 30+ AI projects, obtained a budget of \$700+k over a year
- Built and optimized inference of LLM agents for a startup and for an insurance company
- Gave a [presentation](#) about Adversarial ML at the Google headquarters in Paris.

03/2023 – 09/2024

DATA SCIENTIST, MGH INSTITUTE OF HEALTH PROFESSIONS

Within the SPaN-AD and the Speech and Feeding Disorders Laboratories, I worked on building the bridge between speech and illnesses such as **dementia, ALS or autism**. The research involves computer vision, signal processing and NLP, as well as general data science, Machine Learning and publication skills. I have been accustomed to working with MRI structural and functional data, EEG, Acoustic and kinematic signal, surveys, and more. I am accredited to operate MRIs and EEGs, recruit participants and oversee Research Assistants and Master's student's projects. Speech sciences have taught me great critical thinking in approaching a complex Data Science problem with little data and various modalities, noisy labels and multicollinearity.

Worked on 1 successful grant application (\$250k) and 2 posters.

[Automatic Speech Recognition](#) (ASHA'24, co-authored with Google Euphonia team).

08/2022 – 12/2022

SCIENTIFIC INTERN, SECURITY AND TRUST, UNIVERSITY OF LUXEMBURG

[Constrained adaptative attacks](#) and defenses from computer vision to tabular data with additional domain constraints. Research on data augmentation and robustification methods for deep learning model generalization and reliability.

04/2022 – 08/2022

SCIENTIFIC INTERN, INSTITUTE OF SCIENCE AND TECHNOLOGY AUSTRIA

Building the theoretical framework for the game theory analysis of permissionless Blockchains and enabling fast, asynchronous algorithms with synchronous safety guaranties

[Mandator and Sporades](#) and RACS and SADL: Building a high-performance, asynchrony-resilient algorithm for data geo-replication across data-centers (IEEE Cloud'25)

[Divide & Scale](#): Formalization and roadmap to robust sharding: mathematical study of scaling laws under probabilistic security constraints (SIROCCO'23)

09/2021 – TODAY

CEO, DALEMBERT AI

Co-founder of a **data brokering** Start-up. Hands-on experience on data collection, cleaning and annotation, UX/UI design, pitching, Market Analysis, Scrum/Agile management. Won 2 grants.

02/2021 – 08/2021

MACHINE LEARNING INTERN, SINEQUA

Creation of an **NLP** model for Key Phrase Extraction using BERT for a top-tier Search Engine. Achieved above-State-of-the-Art performance on relevant benchmark using meta-learning techniques; developed multilingual models and studied the effect of model downscaling.

2021 – 2024

FREELANCE CONSULTANT,

5 different missions for industrial companies and scientific startups including healthcare and water management. They involved strategy, causal machine learning and coding.

06/2020 – 12/2020

COLLABORATOR, ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE (EPFL)

Design, proof and optimization of a Leaderless Byzantine fault-tolerant Consensus algorithm for secure Blockchain applications

Publication of [Leaderless Consensus](#), (Best Paper Award at ICDCS'21)

04/2018 – 04/2021

CCO, CODEX

At Codex, an e-tourism start-up I co-founded, I managed the writing of creative content for escape games, marketing, communication and contracts negotiation. Won 7 awards.

01/2020 – 06/2022

TEACHER ASSISTANT, ÉDUCATION NATIONALE

Correcting written IT exams and running weekly English oral exams for engineering students.

EDUCATION

09/2018 – 12/2022

ENGINEERING DEGREE, ÉCOLE DES PONTS PARISTECH - 3.87 GPA

Machine Learning, Statistics, Fourier Analysis, Computer Vision, Stochastic Processes, Game Theory, Object-oriented Programming, Operations Research, web development

- **Google HashCode** Hackathon: top 100 France 2019: score optimization through insertion sorting on a scheduling problem (2019,2020 and 2022 editions)
- **Deep Compress Sensing** for holographic radar in partnership with Airbus
- **KIRO** – Operations Research Hackathon from Renault to optimize trucks delivery routing.
- Various projects involving **simulated annealing**, **genetic algorithms** and **matching algorithms**

09/2021 – 10/2022

MASTER OF RESEARCH, INSTITUT POLYTECHNIQUE DE PARIS - MPRO

Graph theory, Mathematical Programming, Constraint Programming, Exact and Approximate Complexity Theory, Meta-Heuristics, Scheduling, Stochastic Optimization, Queue Theory, OR for Transports and for Sustainable Development, OR and Machine Learning, Localization models

09/2021 - 06/2022

BACHELOR'S DEGREE, UNIVERSITÉ PARIS NANTERRE

Bachelor of Philosophy pursued concurrently with the engineering diploma. Courses include formal logic, philosophy of mind and philosophy of science. Graduated with honors.

09/2015 - 06/2018

PREPARATORY SCHOOL, LYCÉE FÉNELON

PCSI, PC* (Mathematics, Physics, chemistry)

2016 International Chemistry Olympiads (French pre-selection), silver mention

SKILLS

- Python (deep learning libraries)
- GCP Professional ML Engineer Certificate
- AWS ML Specialty Certificate
- C++ (OOP, image vision and alteration)
- Julia (Gurobi, CPLEX, GLPK), VBA
- Basic Knowledge of R, OPL, Solidity, HTML/CSS
- Git, LaTeX, CI/CD, Docker, MLOps
- CITI-certified, EEG/MRI practice and software
- French (native)
- English (fluent)
- German (conversational)
- Japanese (beginner)
- Spanish (beginner)

ACTIVITIES

- Participated in two national debating competitions (in French and in English)
- Founded and directed Philosophy club of Ecole des Ponts ParisTech
- Sports include tennis (11 years in club), skiing, martial arts, biking, hiking
- Handimangement certificate